DoW Agentic Process Flow — DoDAF 2.0

This deck illustrates the full agentic workflow and clarifies the role of each stage in the DoDAF 2.0 process.

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Document Revision History

Date	Author	Description	Version
11/15/2025	Ravi Natarajan	Initial Draft	V1.0
11/19/2025	Ravi Natarajan	1. Added HIL (yellow rect) in the main flow 2. Added a slide to show how artifacts are collected, produced for final report 3. Added acronyms page	V1.1
11/21/2025	Ravi Natarajan	Added 1 slide after Intro slide to show the Main Agentic Process flow pipelines – which comprise of DoDAF 2.0, JCDIS, DaU SE pipelines. Ensured, to add WKC (Watsonx Knowledge Catalog) which is the vital component for the DoW Retriever.	V1.2
11/30/2025	Ravi Natarajan	Ensured that the Knowledge Catalog is generic and need not be WKC but any other product as suitable.	V1.3

Introduction

Purpose of This Deck

- Explain the DoW Agentic Process Flow aligned to DoDAF 2.0
- Describe how uploaded documents move through the 6 automated stages
- Show how supporting components (Docling, Retriever, Neo4J, Rhapsody, PostgreSQL) fit together
- Provide a coherent end-to-end view of the system architecture

What This Deck Covers

- Full agentic workflow overview
- Stage-by-stage walkthrough (1–6)
- Supporting subsystem roles
- How each component contributes to DoDAF analysis and report generation
- slide (4) “Main Agents Process Flow” shows the full flow at high level
- slide (20) “DoDAF 6-Stage CrewAI Pipeline” shows crews and list of agents within each crew

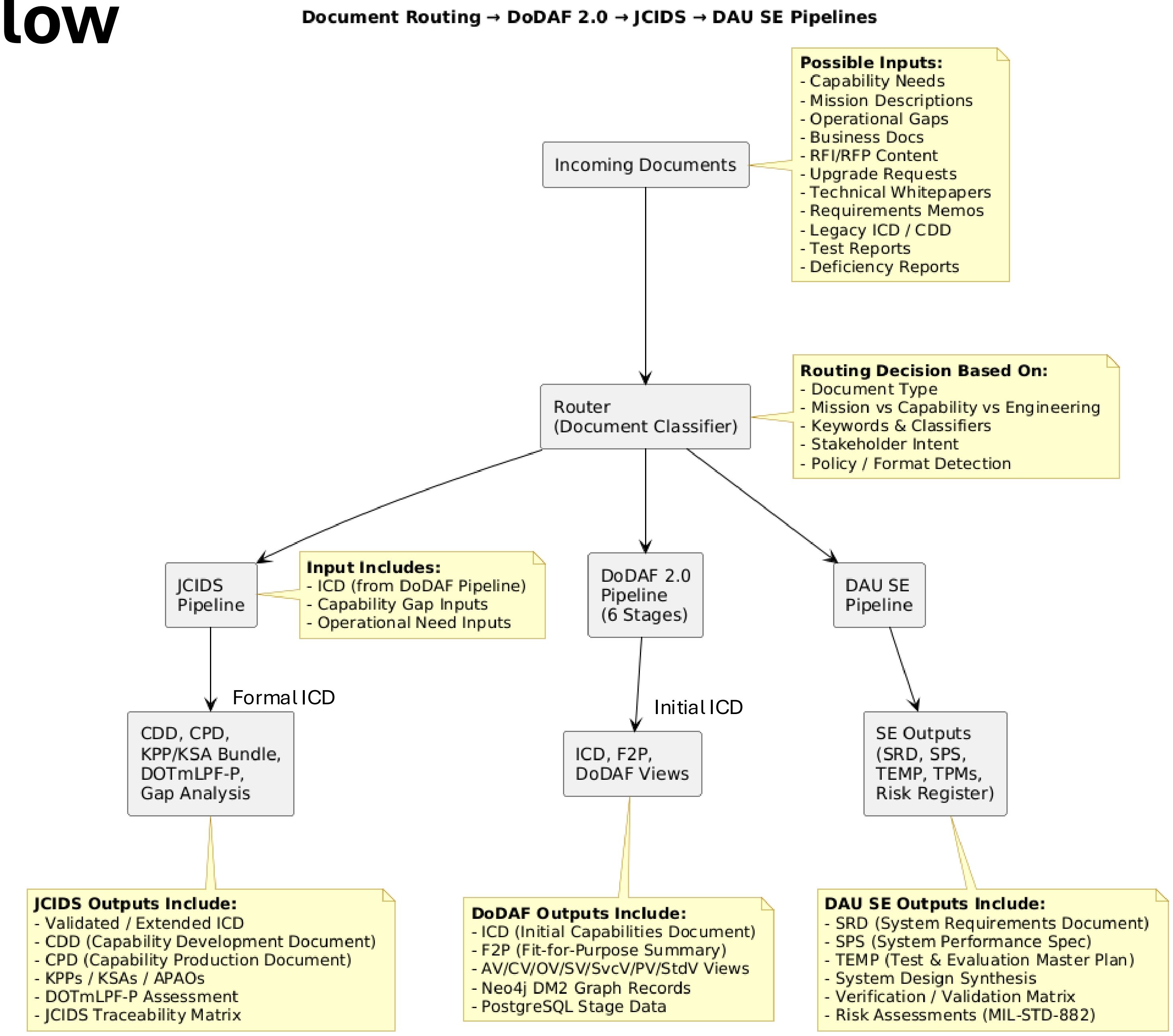
Outcome

- A clear understanding of how the DoW pipeline automates DoDAF 2.0 architectural processing.

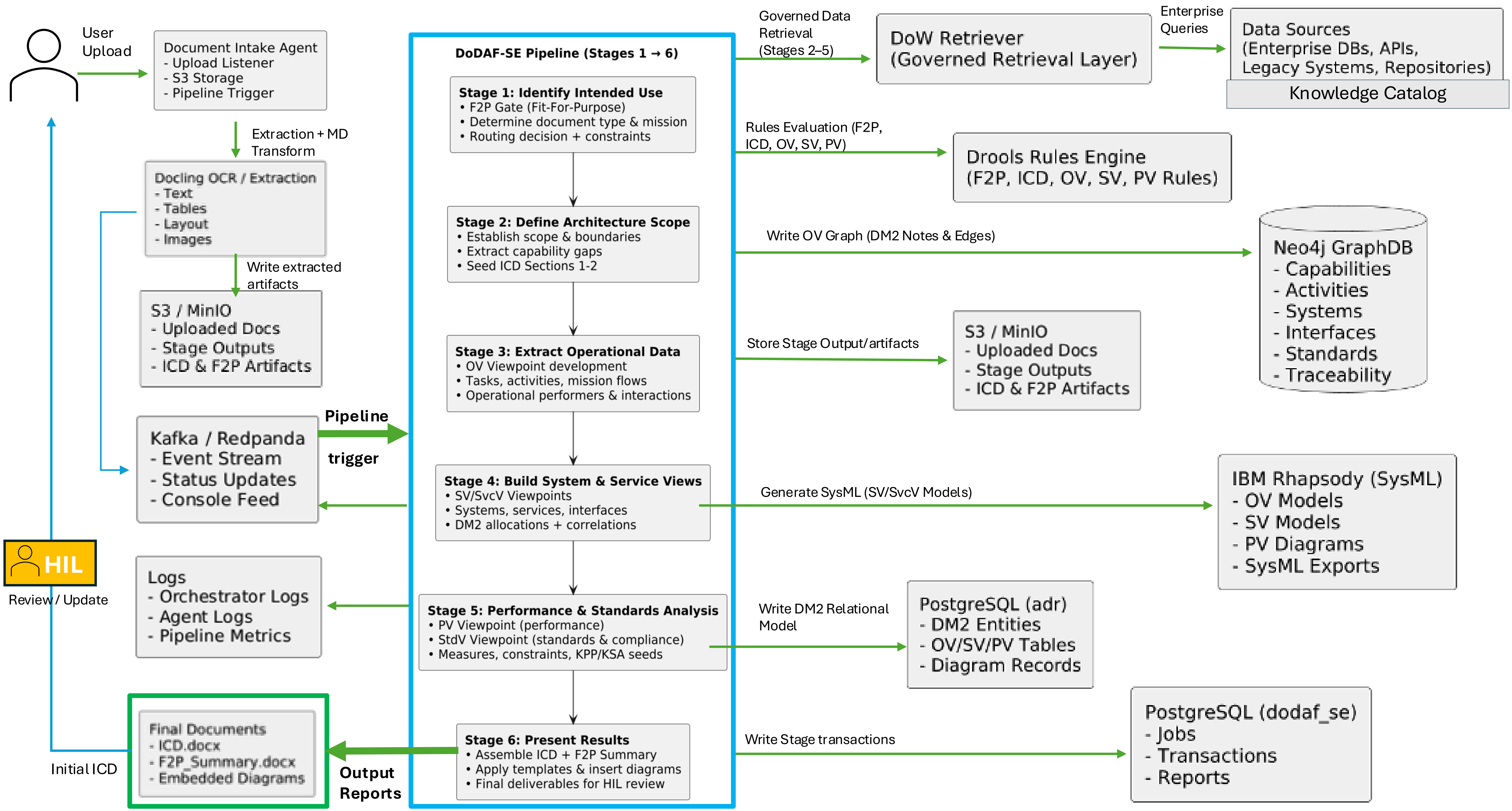
Main Agentic Process Flow

1. All incoming documents are classified by a Router to determine the correct processing path.
2. The DoDAF 2.0 pipeline executes first when architectural decomposition or mission/context analysis is required.
3. DoDAF 2.0 pipeline outputs include the ICD, F2P summary, and full OV/CV/SV/PV/StdV architectural views.
4. The initial ICD (HIL-approved) is used as key input to the JCIDS pipeline for capability development.
5. The JCIDS pipeline produces CDD, CPD, KPP/KSA bundles, DOTmLPF-P assessment, and traceability artifacts.
6. The DAU Systems Engineering pipeline produces SRD, SPS, TEMP, TPMs, and engineering lifecycle deliverables.
7. Each pipeline independently outputs its own formal documents for decision-making.
8. The combined structure ensures traceable flow from architecture → capability → systems engineering.

The DoW Retriever (not shown in the diagram leverages WKC's (or similar product) governed metadata and lineage to deliver rich, trusted context to all agents.

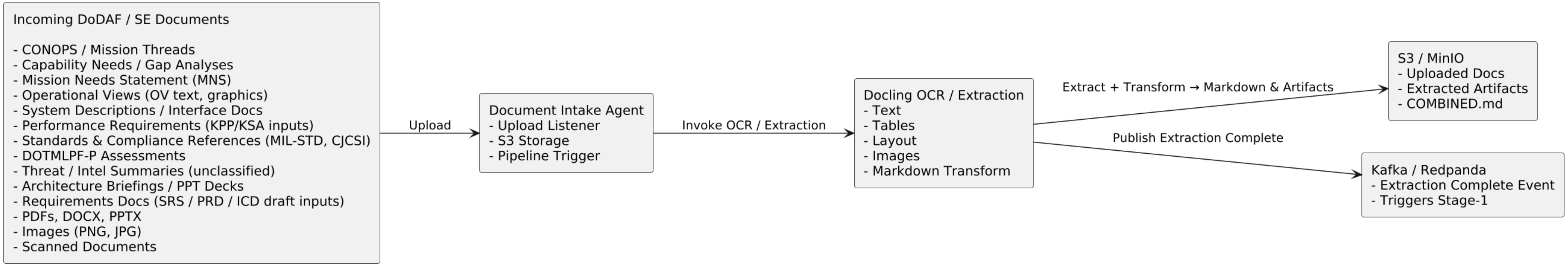


Note: This PPT Deck only shows the DoDAF 2.0 6 stages flow



User Upload
Document Intake Agent
- Upload Listener
- S3 Storage
- Pipeline Trigger
Extraction + MD Transform
Docling OCR / Extraction
- Text
- Tables
- Layout
- Images
Write extracted artifacts
S3 / MinIO
- Uploaded Docs
- Stage Outputs
- ICD & F2P Artifacts
Kafka / Redpanda
- Event Stream
- Status Updates
- Console Feed
Pipeline trigger
HIL
Review / Update
Logs
- Orchestrator Logs
- Agent Logs
- Pipeline Metrics
Initial ICD
Final Documents
- ICD.docx
- F2P_Summary.docx
- Embedded Diagrams
Output Reports
DoDAF-SE Pipeline (Stages 1 → 6)
Stage 1: Identify Intended Use
• F2P Gate (Fit-For-Purpose)
• Determine document type & mission
• Routing decision + constraints
Stage 2: Define Architecture Scope
• Establish scope & boundaries
• Extract capability gaps
• Seed ICD Sections 1-2
Stage 3: Extract Operational Data
• OV Viewpoint development
• Tasks, activities, mission flows
• Operational performers & interactions
Stage 4: Build System & Service Views
• SV/SvcV Viewpoints
• Systems, services, interfaces
• DM2 allocations + correlations
Stage 5: Performance & Standards Analysis
• PV Viewpoint (performance)
• StdV Viewpoint (standards & compliance)
• Measures, constraints, KPP/KSA seeds
Stage 6: Present Results
• Assemble ICD + F2P Summary
• Apply templates & insert diagrams
• Final deliverables for HIL review
Governed Data Retrieval (Stages 2–5)
DoW Retriever (Governed Retrieval Layer)
Enterprise Queries
Data Sources (Enterprise DBs, APIs, Legacy Systems, Repositories)
Knowledge Catalog
Rules Evaluation (F2P, ICD, OV, SV, PV)
Drools Rules Engine (F2P, ICD, OV, SV, PV Rules)
Write OV Graph (DM2 Notes & Edges)
Neo4j GraphDB
- Capabilities
- Activities
- Systems
- Interfaces
- Standards
- Traceability
Store Stage Output/artifacts
S3 / MinIO
- Uploaded Docs
- Stage Outputs
- ICD & F2P Artifacts
Generate SysML (SV/SvcV Models)
IBM Rhapsody (SysML)
- OV Models
- SV Models
- PV Diagrams
- SysML Exports
Write DM2 Relational Model
PostgreSQL (adr)
- DM2 Entities
- OV/SV/PV Tables
- Diagram Records
Write Stage transactions
PostgreSQL (dodaf_se)
- Jobs
- Transactions
- Reports

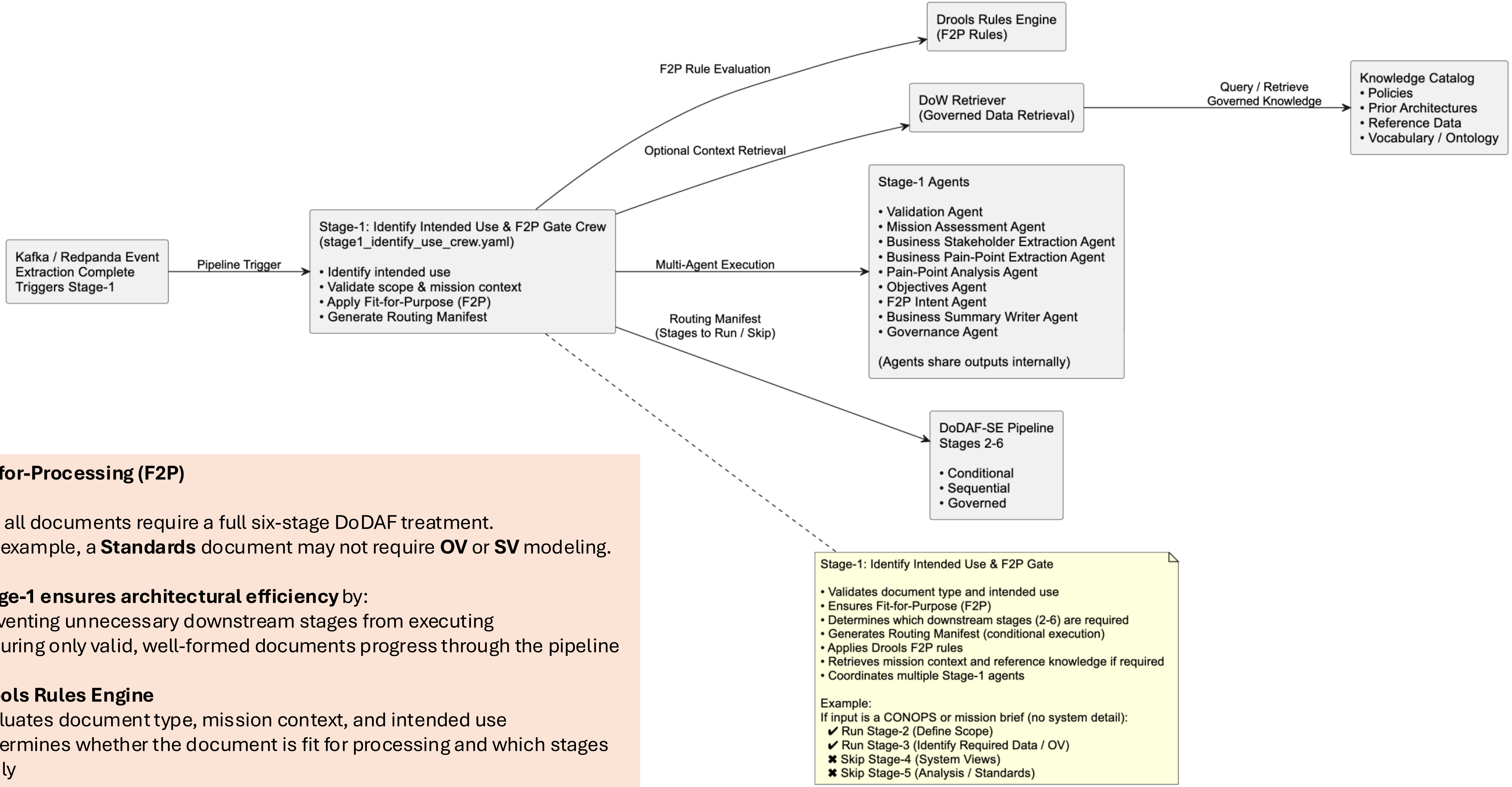
Incoming Documents Flow
Intake & Pre-Processing



Document Intake Agent:
- Listens for uploads
- Places documents into S3/MinIO storage
- Sends a pipeline trigger event to Kafka/Redpanda

Markdown output includes:
- Extracted text
- Hierarchical headings
- Tables translated into Markdown table syntax
- Bullet lists and callouts
- Embedded image references
- Metadata (page number, confidence scores, bounding boxes)

Pipeline Trigger & Stage-1 (F2P Gate)



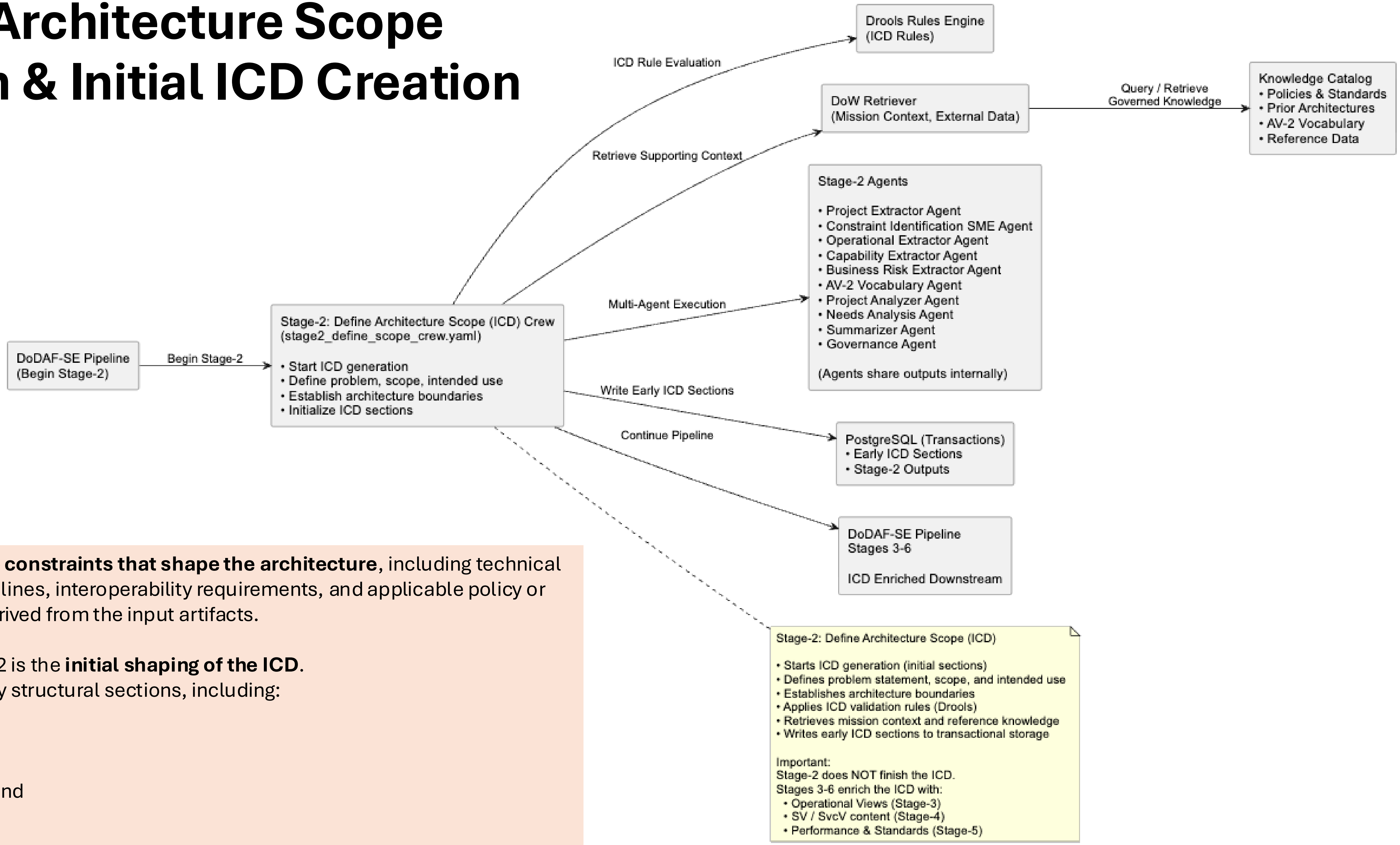
Fit-for-Processing (F2P)

Not all documents require a full six-stage DoDAF treatment.
For example, a **Standards** document may not require **OV** or **SV** modeling.

Stage-1 ensures architectural efficiency by:
Preventing unnecessary downstream stages from executing
Ensuring only valid, well-formed documents progress through the pipeline

Drools Rules Engine
Evaluates document type, mission context, and intended use
Determines whether the document is fit for processing and which stages apply

Stage-2: Architecture Scope Definition & Initial ICD Creation

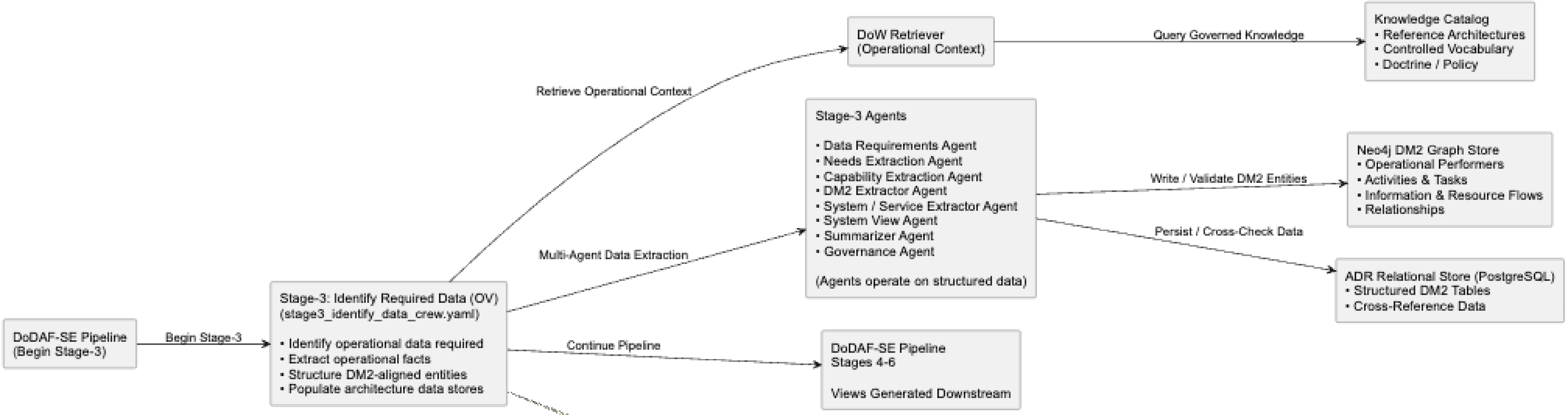


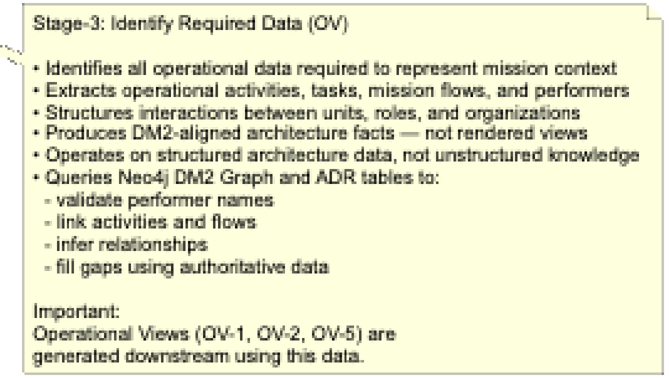
Stage-2 establishes the constraints that shape the architecture, including technical limitations, mission timelines, interoperability requirements, and applicable policy or standards references derived from the input artifacts.

A key outcome of Stage-2 is the **initial shaping of the ICD**.
Agents generate the early structural sections, including:
high-level purpose,
mission context,
capability gaps,
stakeholder groupings, and
the scope statement.

This forms the **architectural foundation** that subsequent stages—**OV, SV, SvcV, and PV**—progressively enrich and complete.

Stage-3: Identify & Structure Operational Data



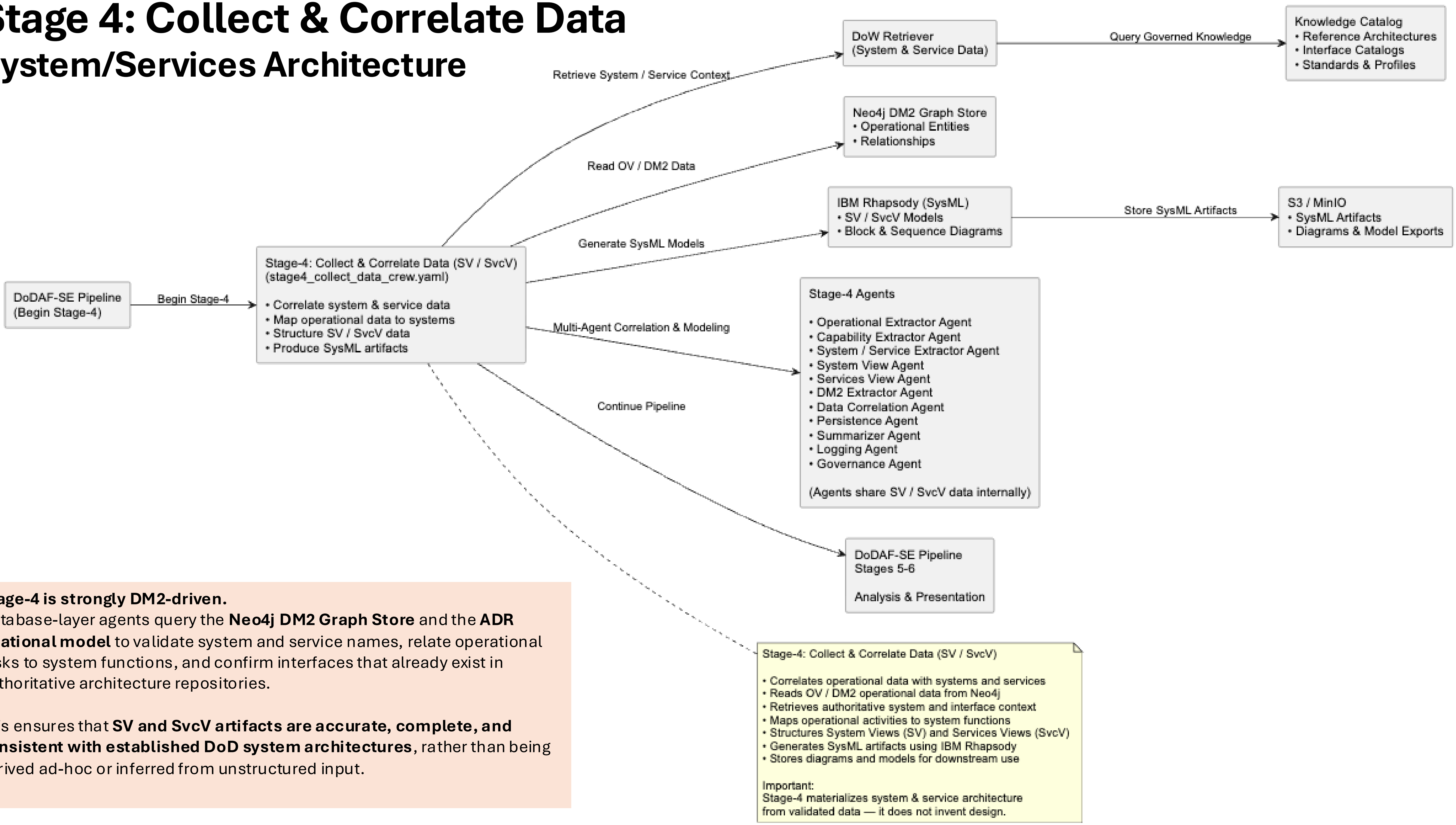


The goal of Stage-3 is to identify and structure all operational data required to accurately represent the mission context.
Stage-3 does **not** generate views; it produces the **data foundation** used to construct views downstream.

This includes extracting operational activities, tasks, mission flows, operational performers (units, roles, organizations), and the interactions between them.

A key distinction is that **Stage-3 operates on structured architectural facts, not unstructured knowledge**.
Database-layer agents may query the **DM2 Graph Store (Neo4j)** or **ADR relational tables (PostgreSQL)** to validate performer names, link activities, infer relationships, and fill gaps using authoritative DM2-aligned data.

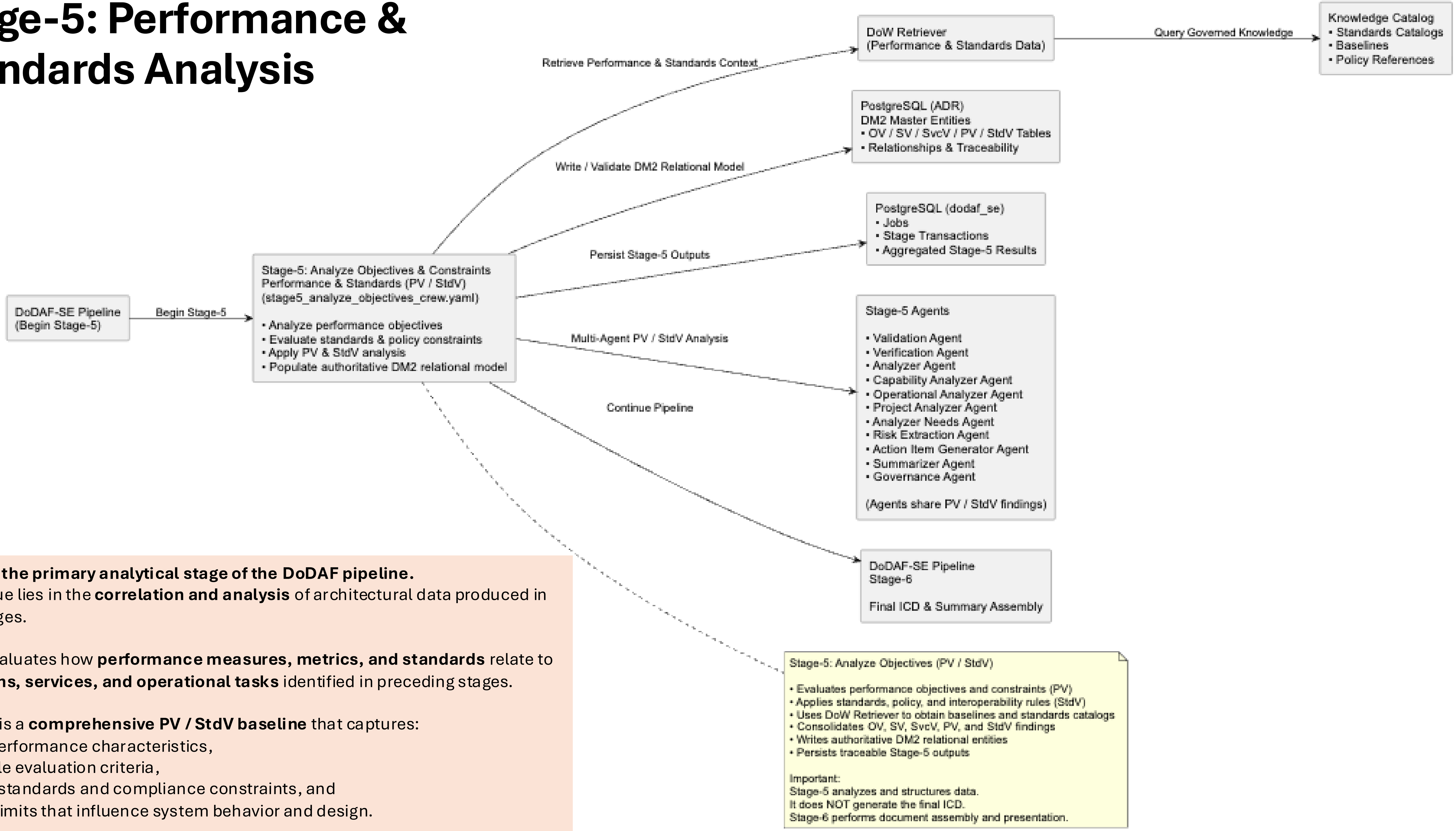
Stage 4: Collect & Correlate Data
System/Services Architecture



Stage-4 is strongly DM2-driven.
Database-layer agents query the **Neo4j DM2 Graph Store** and the **ADR relational model** to validate system and service names, relate operational tasks to system functions, and confirm interfaces that already exist in authoritative architecture repositories.

This ensures that **SV and SvcV artifacts are accurate, complete, and consistent with established DoD system architectures**, rather than being derived ad-hoc or inferred from unstructured input.

Stage-5: Performance & Standards Analysis

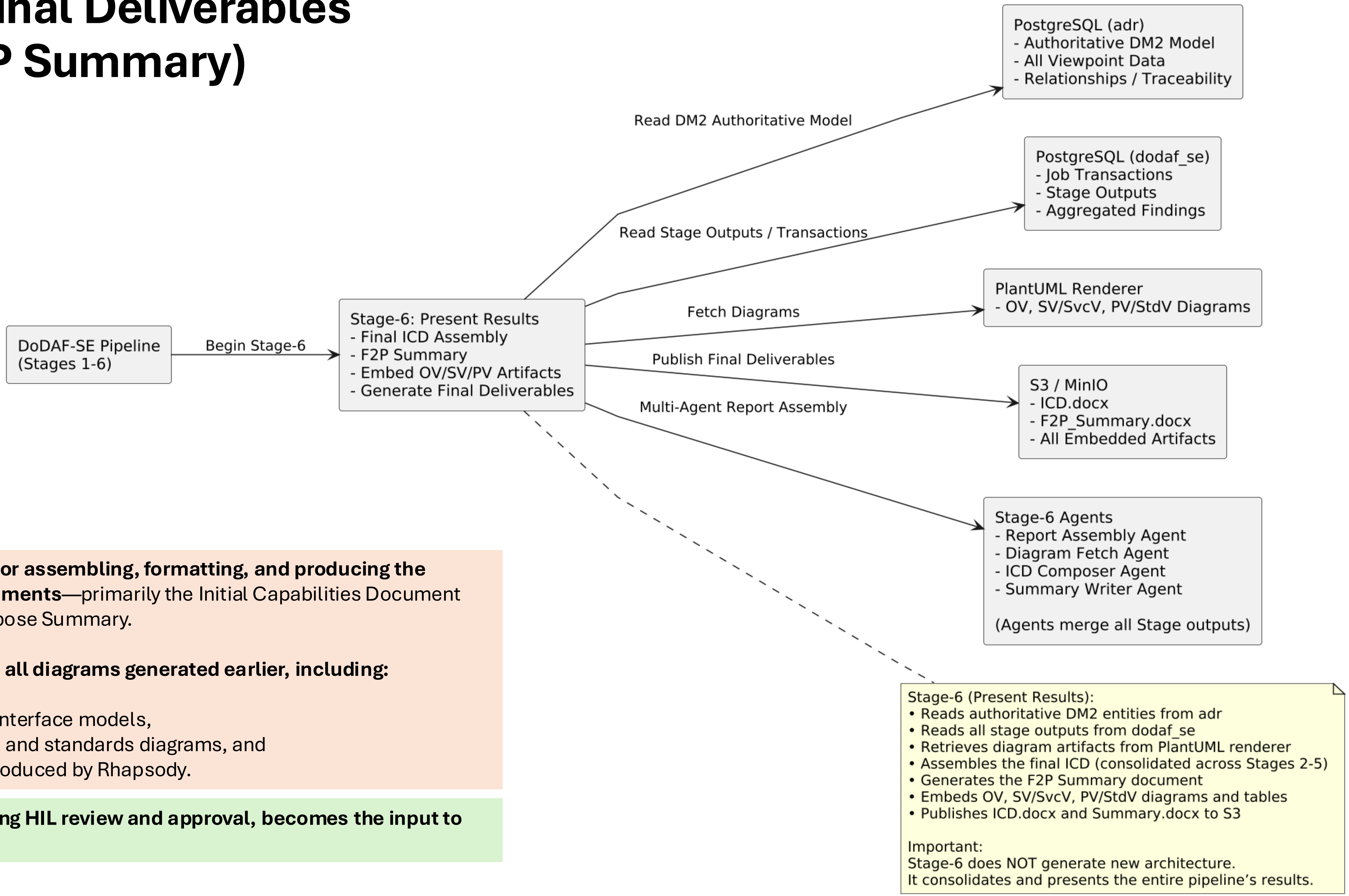


Stage-5 is the primary analytical stage of the DoDAF pipeline.
Its key value lies in the **correlation and analysis** of architectural data produced in earlier stages.

Stage-5 evaluates how **performance measures, metrics, and standards** relate to the **systems, services, and operational tasks** identified in preceding stages.

The result is a **comprehensive PV / StdV baseline** that captures:
required performance characteristics,
measurable evaluation criteria,
governing standards and compliance constraints, and
technical limits that influence system behavior and design.

Stage-6: Final Deliverables (ICD & F2P Summary)



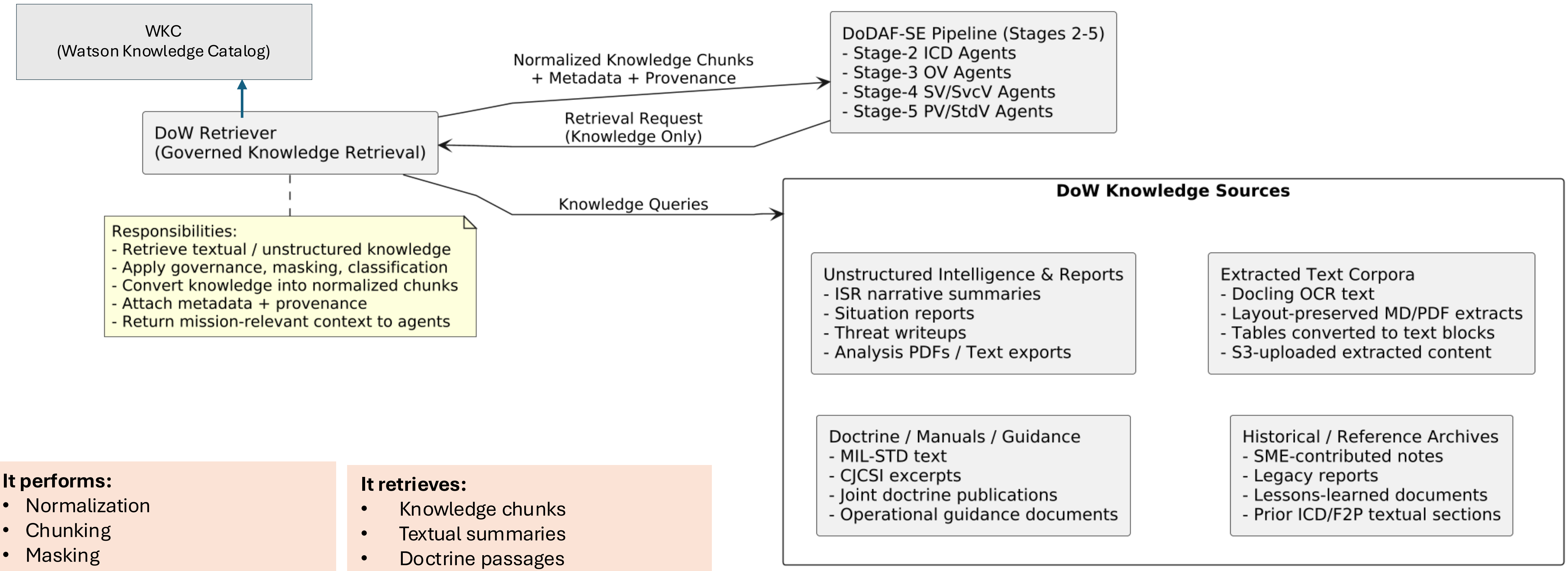
Stage-6 is responsible for assembling, formatting, and producing the DoDAF-compliant documents—primarily the Initial Capabilities Document (ICD) and the Fit-for-Purpose Summary.

This stage also embeds all diagrams generated earlier, including:

- **OV** operational flows,
- **SV/SvcV** system and interface models,
- **PV/StdV** performance and standards diagrams, and
- any **SysML exports** produced by Rhapsody.

This Output upon passing HIL review and approval, becomes the input to the JCIDS Pipeline.

Enterprise Retrieval Layer for DoW Architecture Data



It performs:
- Normalization
- Chunking
- Masking
- Governance
- Provenance scoring
- Knowledge fusion from multiple knowledge sources

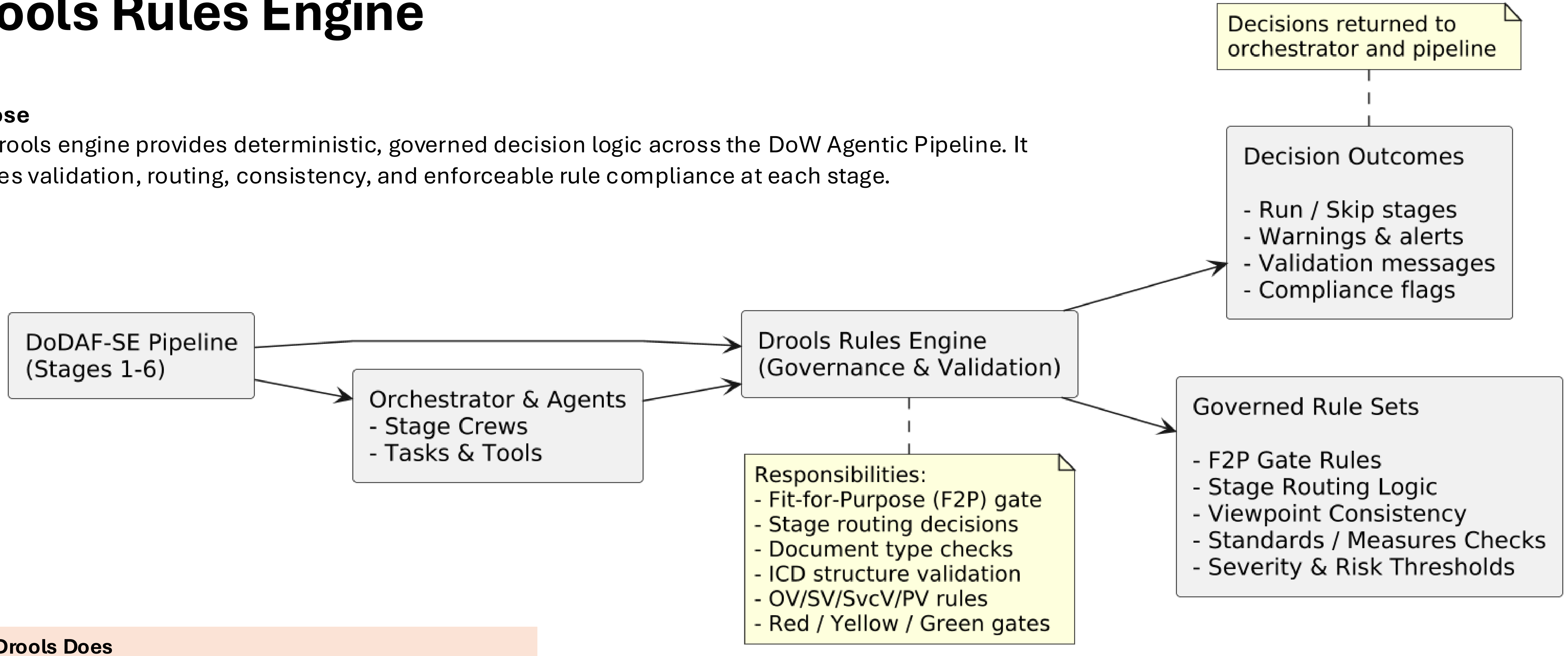
It retrieves:
- Knowledge chunks
- Textual summaries
- Doctrine passages
- Unstructured intel
- Notes / reports / PDFs converted by Docling
- SME-written content
- Historical documents
- Reference knowledge

Data Sources shown are examples

Drools Rules Engine

Purpose

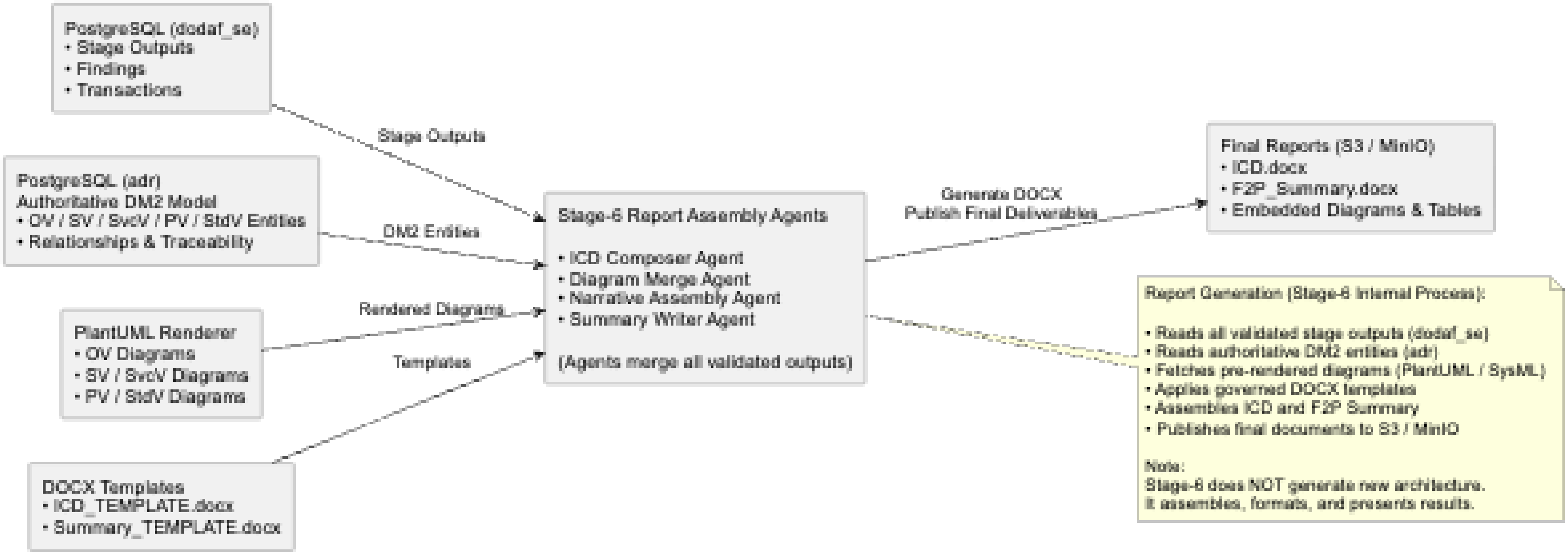
The Drools engine provides deterministic, governed decision logic across the DoW Agentic Pipeline. It ensures validation, routing, consistency, and enforceable rule compliance at each stage.



What Drools Does

- Enforces **F2P (Fit-for-Purpose) Gate rules**
- Validates **document type, scope, and intended use**
- Determines required **downstream stages (2–6)**
- Applies **DoDAF architectural constraints**
- Provides **ICD structural checks**
- Governs **OV/SV/SvcV/PV rule logic**
- Supports **red/yellow/green gating decisions**
- Ensures **governed, repeatable outputs**

Report Generation Process



The Report Assembly Agents retrieve structured data from PostgreSQL (dodaf_se and adr), merge it with diagrams produced by PlantUML and Rhapsody, and populate the ICD and Fit-for-Purpose templates.

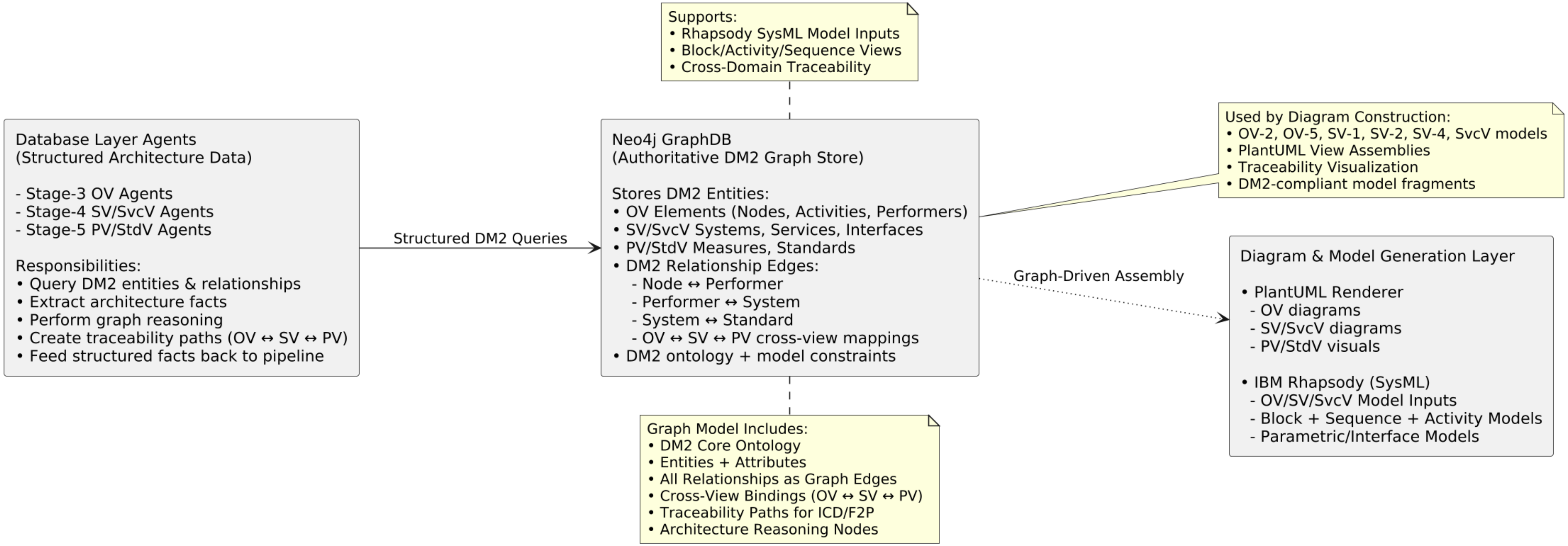
Report Output:
The result is a **fully generated, publication-ready ICD and Fit-for-Purpose Summary**, including:
structured architecture sections synthesized across all stages,
embedded **OV, SV, SvcV, PV, and StdV** diagrams,
end-to-end traceability with **DM2-aligned references**, and
consistent formatting with **governance-aligned content**.

Artifacts produced at various stages and final Report

Stage	Primary Function	DoDAF Views Produced / Informed	Exportable?	Used In Final Products
Stage-1	Identify Intended Use (F2P Gate)	No DoDAF views generated (routing + classification only)	N/A	Provides routing + purpose + constraints for ICD
Stage-2	Define Architecture Scope (ICD Framing)	**CV Inputs**:• CV-1 (Vision) — partially inferable; SME/HIL needed CV-2 (capability taxonomy seed)• CV-3 (initial dependencies)Not exported, but internally produced as text	Partial	Creates **ICD Sections 1–2** foundation
Stage-3	Operational Viewpoint Extraction	**DoDAF Views Produced / Informed:** Extract and structure all operational-level data required to inform Operational Views (OVs). *Stage-3 does not generate diagrams*. Provides the curated operational dataset used downstream to produce OV-1/2/3/5/6 (via PlantUML/Rhapsody generators).	**Yes** (via PlantUML)	Populates **ICD operational context**, missions, activities, resource flows, constraints, and gaps.
Stage-4	System & Service Viewpoint Modeling	**SV/SvcV Views**:✓ SV-1 (Systems/Services)✓ SV-2 (Interfaces)✓ SV-4 (Functions)✓ SvcV-1/2/3 equivalents✓ SysML diagrams (via Rhapsody), Allocations.	**Yes** (PlantUML + Rhapsody exports)	Populates **ICD architecture sections + diagrams**
Stage-5	Performance / Standards Analysis	**PV/StdV Views**:✓ PV-1 (Performance Parameters)✓ PV-2 (Performance Constraints)✓ StdV-1 (Standards Profile)✓ StdV-2 (Standards Forecast)	Yes (tables/graphs)	Populates **ICD KPP/KSAs, standards, technical constraints**
Stage-6	Present Results / Document Assembly	No new DoDAF views — assembles everything	N/A	**Generates → JOBID_ICD.docx + JOBID_F2P_Summary.docx**

Final Deliverables (User-Facing): JOBID_ICD.docx, JOBID_F2P_Summary.docx

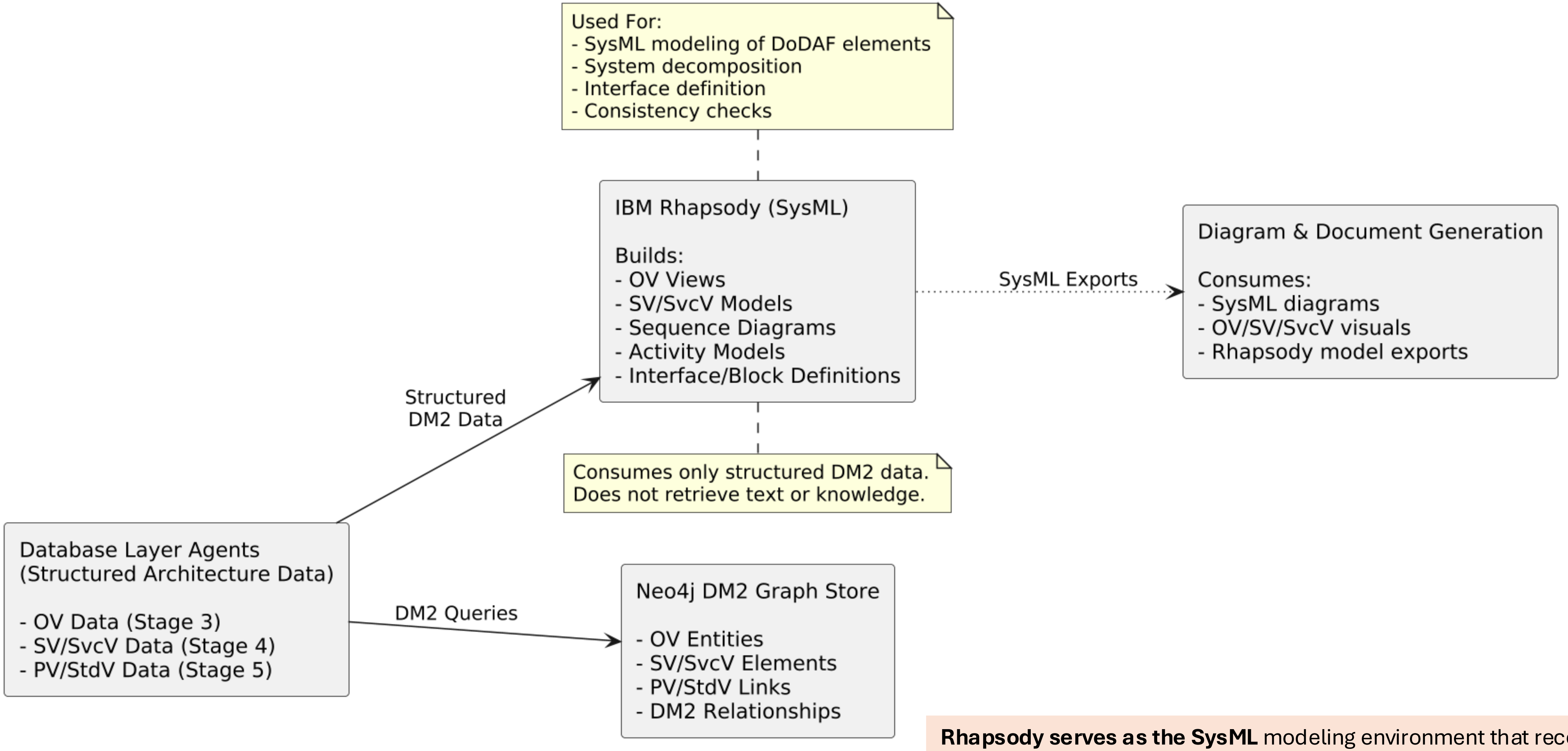
Neo4J GraphDB



When stages 3 through 5 run, they populate Neo4J with:
Operational View entities such as activities, performers, nodes, and exchanges
System and Service View elements, including systems, interfaces, ports, services, and resource flows
Performance and Standards View relationships, such as measures, rules, constraints, and standards applicability

The real power comes from the relationships: OV → SV/SvcV → PV/StdV.
These allow agents to ask questions like:
"Which system supports this operational activity?"
"Which standards constrain this interface?"
"What is the traceability path from an OV mission objective down to a PV performance measure?"

IBM Rhapsody



Rhapsody serves as the SysML modeling environment that receives structured architectural content from earlier stages and transforms it into standards-compliant systems engineering models.
While Neo4J provides semantic relationships and PlantUML generates quick visual diagrams, Rhapsody produces the formal SysML artifacts expected in larger DoD programs.

Components and brief description

Technology / Product	Role in the Architecture
Docling	OCR + layout extraction of PDFs/Word/images into structured Markdown and text blocks.
Drools Rules Engine	Executes Fit-for-Purpose rules, routing logic, architecture constraints, and validation.
Kafka	Event bus and control plane: triggers stages, streams logs, and synchronizes workflow execution. Pipelines get triggered by events. eg. JCIDS pipeline gets triggered when an event is published to the jcids_in topic with link to input artifacts in the S3 bucket.
S3	Object storage for uploaded documents, extracted text, diagrams, and final ICD/F2P outputs.
Neo4J GraphDB	DM2-based knowledge graph storing OV/SV/SvcV/PV entities and relationships; supports traceability.
IBM Rhapsody	SysML modeling environment for OV/SV/SvcV/PV views; generates formal engineering diagrams.
PostgreSQL	Structured data storage for stage outputs (dodaf_se) and authoritative DM2 entities (adr).
Knowledge Catalog	Example: WKC unifies business and technical stewardship in one governed platform, enabling our DoW Retriever to provide agents with rich, trusted context. Similar products like WKC as suitable for FedRAMP env.
FastAPI	Backend service layer for orchestrator APIs, job submission, status, console streaming, and report retrieval.

WKC ensures every component operates on governed, trusted, and well-stewarded data—dramatically improving the quality and reliability of agent outputs.
The DoW Retriever leverages WKC's governed metadata and lineage to deliver rich, trusted context to all agents.

DoDAF 6-Stage CrewAI Pipeline - Crews & Agents

Stage	Agents
stage1_identify_use_crew	validation_agent, mission_assessment_agent, bd_stakeholder_extractor_agent, bd_pain_point_extractor_agent, pain_point_agent, objectives_agent, f2p_intent_agent, bd_summary_writer_agent, governance_agent
Stage2_define_scope_crew	project_extractor_agent, constraint_identification_sme_agent, operational_extractor_agent, capability_extractor_agent, bd_risk_extractor_agent, av2_vocabulary_agent, project_analyzer_agent, analyzer_needs_agent, summarizer_agent, governance_agent
stage3_required_data_crew	data_requirements_agent, extractor_needs_agent, capability_extractor_agent, dm2_extractor_agent, system_service_extractor_agent, system_view_agent, summarizer_agent, governance_agent
stage4_collect_correlate_data_crew	operational_extractor_agent, capability_extractor_agent, system_service_extractor_agent, system_view_agent, services_view_agent, dm2_extractor_agent, data_correlation_agent, persistence_agent, summarizer_agent, logging_agent, governance_agent
stage5_analyze_objectives_crew	validation_agent, verification_agent, analyzer_agent, capability_analyzer_agent, operational_analyzer_agent, project_analyzer_agent, analyzer_needs_agent, bd_risk_extractor_agent, bd_actionitem_generator_agent, summarizer_agent, governance_agent

Acronyms

Acronym	Meaning	Acronym	Meaning
DoDAF	Dept. of Defense Architecture Framework	**PV**	Performance Viewpoint
DM2	DoD MetaModel	**StdV**	Standards Viewpoint
ICD	Initial Capabilities Document	**OV-1**	Operational Concept Graphic
F2P	Fit-for-Purpose	**OV-2**	Operational Resource Flow
OV	Operational Viewpoint	**OV-3**	Operational Event Trace
CV	Capability Viewpoint	**OV-5a/b**	Activity Model
SV	Systems Viewpoint	**OV-6c**	State Transition
SvcV	Services Viewpoint	**SV-1**	Systems Interface
PV-1	Performance Parameters	**SV-2**	Systems Resource Flow
PV-2	Performance Constraints	**SV-4**	Systems Functionality & Allocations
SvcV-1/2/3	Service Models / Interfaces / Flows	**StdV-1**	Standards Profile
ADR	Authoritative Data Repository	**StdV-2**	Standards Forecast
ISR	Intelligence, Surveillance & Reconnaissance	**HIL**	Human-In-the-Loop
ORBAT	Order of Battle	**S3**	Object Storage (MinIO)
OCR	Optical Character Recognition	**MCP**	Model Context Protocol
API	Application Programming Interface	**SysML**	Systems Modeling Language

End-to-End DoDAF 2.0 Agentic Workflow - Summary

This architecture delivers an end-to-end, fully agentic DoDAF 2.0 workflow

- Automated intake → OCR → structured extraction
- Agentic six-stage DoDAF pipeline
- Federated knowledge retrieval via DoW Retriever
- knowledge base like WKC the retriever would use to fetch domain specific data
- Graph-based reasoning with Neo4j
- Formal SysML modeling in IBM Rhapsody – or similar products.
- Automated diagram generation (OV/SV/PV)
- Structured storage in PostgreSQL (dodaf_se + adr)
- Automated DOCX report generation (ICD + F2P Summary)
- Complete traceability, provenance, and governance
- Fully orchestrated and repeatable architecture pipeline

Questions?

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